# Probability and Statistics – Elementary Probability Theory

## Random Variables

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# Definition

- A *probability space* $(S, \mathcal{F}, \mathrm{P})$ is a triplet that models our random experiment by means of a probability measure $\mathrm{P}(E)$ defined on subsets $E \subseteq S$ of the sample space $S$ belonging to the $\sigma$-algebra $\mathcal{F}$.

- Within this space, we may want to study quantities of interest that are function of randomly occurring events, e.g., inflation, temperature, exchange rates, job response times, ....

- Random variables provide a formalism to map these variables of interest to numerical values.

# Definition

- A **Random Variable** (r.v.) is a mapping from the sample space to the real numbers. So if $X$ is a random variable, $X : S \to \mathbb{R}$.
- Each element of the sample space $s \in S$ is therefore assigned by $X$ a numerical value $X(s)$.
- If we denote the generic (unknown) outcome of the random experiment as $s$, then the corresponding outcome of the random variable $X(s)$ will be generically referred to as $X$.

# Examples of Random Variables

Consider once again the experiment of rolling a single fair die.

- Then $S = \{\epsdice{1}, \epsdice{2}, \epsdice{3}, \epsdice{4}, \epsdice{5}, \epsdice{6}\}$ and for any $s \in S$, $\mathrm{P}(\{s\}) = \frac{1}{6}$.
- An obvious random variable to define on S is $X : S \to \mathbb{R}$, s.t.

$$
\begin{aligned}
X(\epsdice{1}) &= 1, \\
X(\epsdice{2}) &= 2, \\
&\vdots \\
X(\epsdice{6}) &= 6.
\end{aligned}
$$

- Then e.g. $\mathrm{P}_X(1 < X \leq 5) = \mathrm{P}(\{\epsdice{2}, \epsdice{3}, \epsdice{4}, \epsdice{5}\}) = \frac{4}{6} = \frac{2}{3}$
- and $\mathrm{P}_X(X \in \{2, 4, 6\}) = \mathrm{P}(\{\epsdice{2}, \epsdice{4}, \epsdice{6}\}) = \frac{1}{2}$.

- Alternatively, we could define a random variable $Y : S \to \mathbb{R}$, s.t.

$$Y(\epsdice{1}) = Y(\epsdice{3}) = Y(\epsdice{5}) = 0,$$
$$Y(\epsdice{2}) = Y(\epsdice{4}) = Y(\epsdice{6}) = 1.$$

- Then clearly

$$\mathrm{P}_Y(Y = 0) = \mathrm{P}(\{\epsdice{1}, \epsdice{3}, \epsdice{5}\}) = \frac{1}{2}$$

and

$$\mathrm{P}_Y(Y = 1) = \mathrm{P}(\{\epsdice{2}, \epsdice{4}, \epsdice{6}\}) = \frac{1}{2}.$$

Random variable with a finite set of possible outcomes are called *simple*. In general, they may also be countable, in which case they are called *discrete*, or they may be *continuous*.

# Induced Probability

# Induced Probability

- How can we formalize in general the probability that $X$ assumes a specific value $x$?

- Using the probability measure $\mathrm{P}$ already defined on $S$, we may obtain a new probability function $P_X$ on the random variable $X$ in $\mathbb{R}$ with the following procedure.

- For each $x \in \mathbb{R}$, let $S_x \subseteq S$ be the set containing just those elements of $S$ which are mapped by $X$ to numbers no greater than $x$, i.e., $S_x = \{s \in S | X(s) \leq x\}$.

- Then we write

$$\mathrm{P}_X(X \leq x) \equiv \mathrm{P}(S_x)$$

# Support of a RV

- The *image* of $S$ under $X$ is called the **support** of $X$:

$$\mathrm{supp}(X) \equiv X(S) = \{x \in \mathbb{R} \mid \exists s \in S \text{ s.t. } X(s) = x\}$$

- So as $S$ contains all the possible outcomes of the experiment, $\mathrm{supp}(X)$ contains all the possible outcomes for the random variable $X$.

- Thus, $\mathrm{P}_X(X \leq x)$ is defined for all $x \in \mathrm{supp}(X)$.

# Example

Consider the experiment of tossing a fair coin, with sample space $\{\mathsf{H},\mathsf{T}\}$ and probability measure $\mathrm{P}(\{\mathsf{H}\}) = \mathrm{P}(\{\mathsf{T}\}) = \frac{1}{2}$.

Suppose that we play a betting game where we win 1£ if we get heads, or we lose it otherwise.

- We can define a random variable $X : \{\mathsf{H},\mathsf{T}\} \to \mathbb{R}$ taking values, say,

$$X(\mathsf{T}) = -1,$$
$$X(\mathsf{H}) = 1.$$

- What does $S_x$ look like for some $x \in \mathbb{R}$?

- The set $S_x$ is defined by:

$$S_x = \begin{cases} \emptyset & \text{if } x < -1; \\ \{\mathsf{T}\} & \text{if } -1 \leq x < 1; \\ \{\mathsf{H}, \mathsf{T}\} & \text{if } x \geq 1. \end{cases}$$

- This induces probabilities $P_X$ on $\mathbb{R}$

$$\mathrm{P}_X(X \leq x) = \mathrm{P}(S_x) = \begin{cases} \mathrm{P}(\emptyset) = 0 & \text{if } x < -1; \\ \mathrm{P}(\{\mathsf{T}\}) = \frac{1}{2} & \text{if } -1 \leq x < 1; \\ \mathrm{P}(\{\mathsf{H}, \mathsf{T}\}) = 1 & \text{if } x \geq 1. \end{cases}$$

The key point for the theory is that with our definitions we can now associate a probability to every interval of $\mathbb{R}$, i.e., a random variable $X$ has the intervals $(-\infty, x]$ as its events.

# Cumulative Distribution Function

# Cumulative distribution function

The **cumulative distribution function** (**cdf**) of a random variable $X$, written $F_X(x)$ (or just $F(x)$) is the probability that $X$ takes a value less than or equal to $x$, i.e.,

$$F_X(x) = \mathrm{P}_X(X \leq x)$$

A cdf offers an alternative way to describe the probability measure $\mathrm{P}_X$ for a random variable $X$. It enables a unified treatment of discrete and continuous random variables.

For any random variable $X$, $F_X$ is right-continuous, meaning if a *decreasing* sequence of real numbers $x_1, x_2, \ldots \to x$, then $F_X(x_1), F_X(x_2), \ldots \to F_X(x)$.

# Properties of the cdf

To check a given function, $F_X(x)$, is a valid cdf, we need to verify the following conditions:

1. Monotonicity: $\forall x_1, x_2 \in \mathbb{R}, x_1 < x_2 \Rightarrow F_X(x_1) \leq F_X(x_2)$;
2. $F_X(-\infty) = 0$, $F_X(\infty) = 1$.
3. $F_X$ is right-continuous.

Note that the first two conditions imply $0 \leq F_X(x) \leq 1, \forall x \in \mathbb{R}$.

For finite intervals $(a, b] \subseteq \mathbb{R}$, it is possible to check that

$$\mathrm{P}_X(a < X \leq b) = F_X(b) - F_X(a).$$

This can be done after noting that the event $E = \{X \leq b\}$ may be written as the union $E = H \cup G$ of the disjoint events:

- $H = (-\infty, a]$
- $G = (a, b]$

and the result follows from Axiom 3.

# Comments

- A random variable is simply a numeric relabelling of our underlying sample space, and all probabilities are derived from the associated underlying probability measure.
- Unless there is any ambiguity, we generally suppress the subscript of $\mathrm{P}_X(\cdot)$ in our notation and just write $\mathrm{P}(\cdot)$ for the probability measure associated with a random variable.
- That is, we forget about the underlying sample space and just think about the random variable and its probabilities.
- Essentially, the support of $X$ becomes our sample space $S$.
- Events for random variables are frequently of the kind $X = x$, $X > a$, $X \leq b$, $a \leq X \leq b$ ....